Forechecking Pressure Topology

A Continuous Geometry Framework for Retrieval Denial

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Abstract

Forechecking systems are usually described with discrete labels, but retrieval denial is created by continuous spatial pressure over time. We introduce **Retrieval Denial Topology (RDT)**, a carrier-centric framework that measures pressure intensity, exit-lane availability, 3-second lane closure, and layered support from the first, second, and third forecheckers. Using all 10 Big Data Cup 2026 public games at 5v5, we segment 2,086 defensive-zone retrieval episodes and evaluate 14,921 tracking frames. Higher-pressure retrievals are consistently harder to escape: denial rates rise from 46.1% in the lowest pressure decile to 64.1% in the highest. RDT also predicts defensive-zone turnovers more accurately than simpler proximity-based baselines, improving area under the ROC curve from 0.640 for the nearest-defender model to 0.742 for the best topology model. We further provide coach-native archetypes, score-state splits, and team process benchmarking to support pre-scouting and clip review.

1 Introduction

This paper asks a direct question: **how does forechecking pressure influence defensive-zone retrieval failure, and can a spatial pressure-topology model predict turnovers more accurately than traditional proximity-based approaches?**

Forechecking decisions are often discussed as discrete systems (for example, 1-2-2 or 2-1-2), while execution quality is continuous and timing-dependent. On defensive-zone retrievals, the coaching problem is not only "which system was called," but how quickly viable exit lanes are removed and whether pressure layers are synchronized. Here, **synchronized pressure layers** means the first forechecker pressures the puck immediately, while the second and third forecheckers arrive quickly enough to remove the carrier's next-best outlets before the play can escape. Prior work on zone entries and defensive structure has relied largely on event-level counts; the spatial geometry of pressure arrival is rarely modeled explicitly.

This paper uses a single organizing hypothesis: **retrieval denial is lane-removal dynamics**. We model this dynamic with a carrier-centric pressure topology and evaluate whether it is simultaneously interpretable for coaching and predictive for high-value defensive outcomes. The key claim is that a defender four feet closer to the carrier but with poor skating angle contributes less effective pressure than one six feet away but closing directly into the play, a distinction invisible to count-based approaches.

2 Data and Episode Construction

2.1 Dataset

The analysis uses all 10 Big Data Cup 2026 public games, combining synchronized tracking, event logs, and orientation metadata. The modeling unit is a 5v5 defensive-zone retrieval episode.

2.2 Episode Segmentation and Outcome Definitions

Episodes begin when possession is secured in the defensive zone under 5v5 and end at the first terminal condition: controlled exit allowed, forced dump-out, turnover in defensive zone, stoppage, or timeout guardrail. Segmentation is rule-based and fully auditable.

Table 1: Dataset Scope and Outcome Base Rates

Metric	Value
Episodes	2,086
Tracking frames sampled (stride = 20)	14,921
Episodes with complete role-aware feature set	1,569
Controlled exits allowed	39.9%
Forced dump-outs	28.6%
Turnovers forced in D-zone	26.1%
Stoppage + timeout	5.5%

3 Methods

3.1 Retrieval Denial Topology (RDT)

At each sampled frame, the likely puck carrier is inferred as the nearest offensive skater to puck location. Defender pressure at carrier location x is:

$$P(x) = \sum_i \exp\left(-\frac{\|x - x_i\|^2}{2\sigma_i^2}\right) \exp\left(-\frac{\mathrm{ETA}_i(x)}{\tau}\right).$$

This pressure score is our proposed modeling choice rather than a league-standard formula: it combines distance, estimated arrival time, and skating direction into one continuous measure of how disruptive each defender is to the carrier. The ETA term uses a **bounded-acceleration reachability approximation**, meaning we estimate how fast a defender can realistically reach the carrier if acceleration stays within physically plausible limits. The spread parameter σ_i is direction-dependent: a defender skating toward the carrier has a tighter, more concentrated pressure footprint, while one moving across or away from the play has a broader, weaker footprint. This jointly rewards proximity, arrival speed, and directional commitment; all three are necessary for effective lane denial.

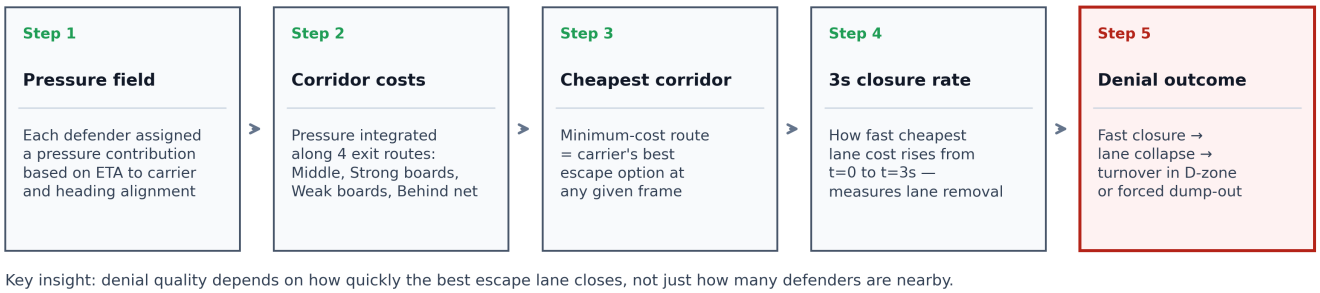

Figure 1: RDT pipeline: each step converts tracking data into an actionable denial signal. The red final box highlights the episode-level outcome that closes the causal chain.

3.2 Corridor Costs and Lane Closure

Four exit corridors are defined: middle lane, strong-side boards, weak-side boards, and behind the net. For each corridor, the cost is computed by integrating the pressure field along the canonical exit path from the carrier's current position. The **cheapest corridor** (the minimum cost across the four options) represents the carrier's best instantaneous escape route. We summarize how pressure evolves with **3-second closure**, defined as cheapest-corridor cost at $t + 3$s minus cheapest-corridor cost at $t = 0$. Positive values mean the best lane became harder to use over the next three seconds; negative values mean the carrier found or preserved an easier exit. We use a 3-second window because it covers the first decisive forecheck wave in most retrievals and keeps the metric anchored to the same early-decision window across episodes.

3.3 Episode-Level Features, Archetypes, and Validation

Frame-level signals are aggregated to episode signatures: mean and peak pressure, corridor costs and closure, compactness, role-geometry distances and pressure shares (F1/F2/F3), D-pinch support count, and score-state context. Here, F1/F2/F3 denote the first, second, and third forecheckers in the pressure sequence. Archetypes are obtained by k-means ($k = 4$) in the pressure-role feature space and labeled by their tactical character. The supervised prediction target is **turnover in defensive zone**, the most valuable denial event because it creates immediate possession gain. Denial rate (turnover + forced dump) is retained as the process lens for coaching-facing interpretation. Validation compares pressure-only and pressure+role models against nearest-distance and defender-count baselines using stratified cross-validated **area under the ROC curve (AUC)**.

4 Results

4.1 Retrieval Geography and Pressure Sanity Check

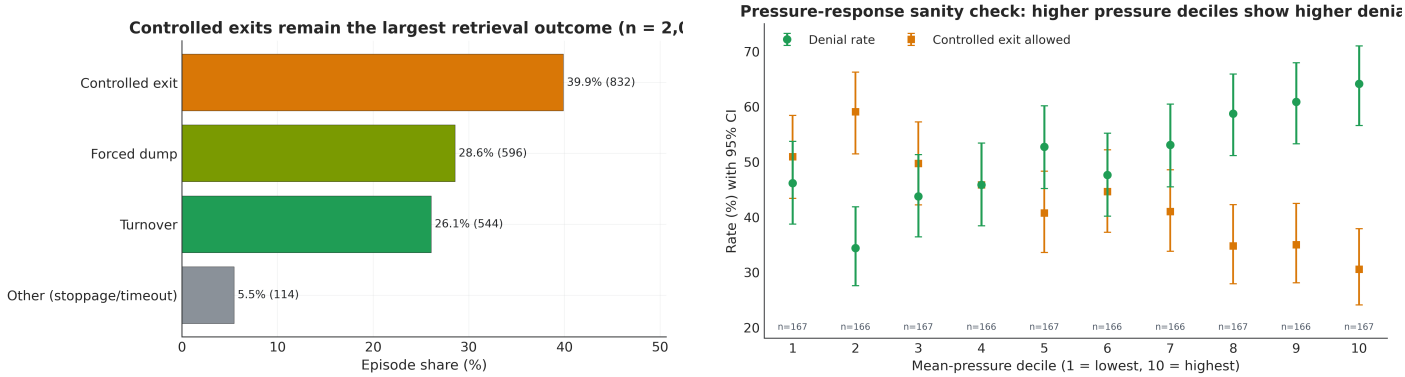

Figure 2: Outcome mix (left) and pressure-response sanity check with decile means and 95% CI (right). Denial rises from 46.1% to 64.1% across pressure deciles 1–10; in other words, high-pressure retrievals are much less likely to escape cleanly.

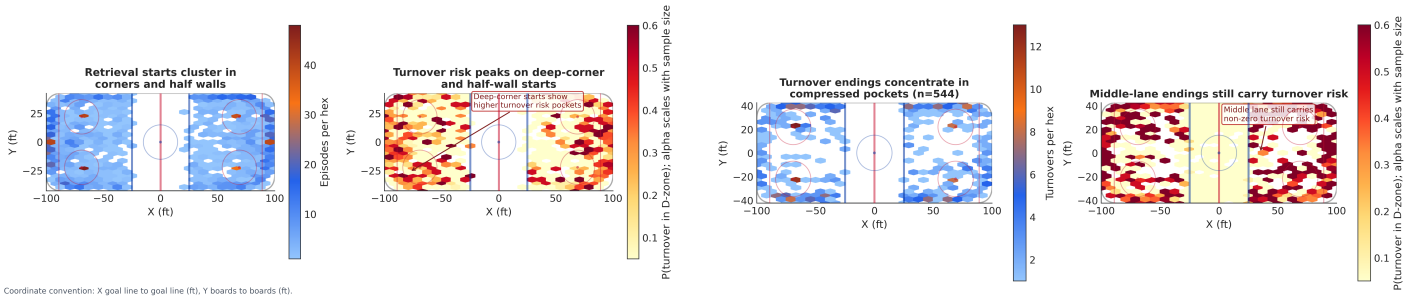

Figure 3: Episode start density and turnover probability by start location (left); turnover end-density and turnover probability by end location (right). Deep-corner starts carry elevated turnover-risk pockets; middle-lane episode endings remain non-zero risk despite the concentration in the deep-corner retrieval space.

Top pressure-quartile denial is 62.6% versus 46.7% in the bottom three quartiles, while controlled exits fall from 46.8% to 32.4%. The monotonic decile response, without the inflection points common to count-based signals, supports pressure quality as a smooth causal pathway to denial rather than a threshold effect.

4.2 Mechanistic Illustration (Single Episode)

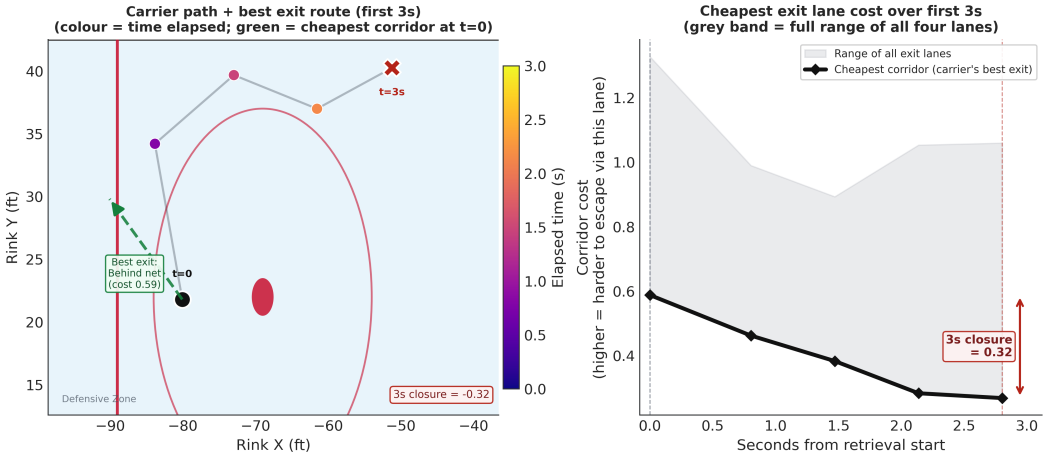

Figure 4: A single illustrative retrieval episode. *Left:* carrier path on the rink section (colour = elapsed time; black circle = t=0 start, red X = t=3s end). The green dashed arrow shows the cheapest available exit corridor at t=0 and its cost; the carrier's actual path diverges from it as defenders close the lane. *Right:* cheapest corridor cost over the 3s window (bold line) with the full range of all four lanes shaded grey. The grey band shows how much the escape options vary within this one episode, not a league-wide trend. The red bracket shows 3-second closure: how much the best exit tightened as defenders converged.

4.3 Archetypes and Relative Topology

Table 2: Coach-Native Archetype Outcome Signatures

Archetype	n	Turnover %	Forced Dump %	Controlled Exit %
Tight Squeeze (C0)	157	33.1	29.3	33.1
Light Pressure (C1)	312	16.7	24.7	52.9
Spread Support (C2)	624	26.1	20.7	45.8
Heavy Disrupt (C3)	476	28.8	27.1	38.2

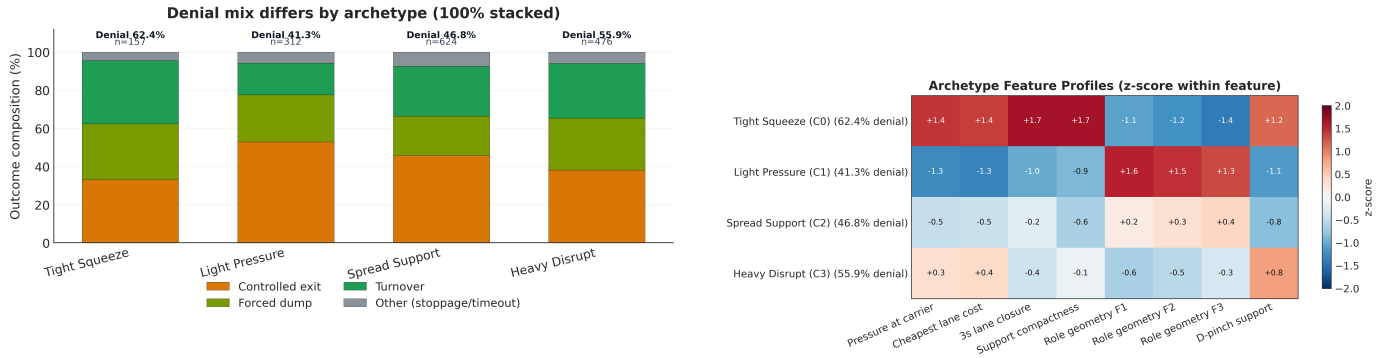

Figure 5: Archetype outcome composition (left, 100% stacked with denial labels) and grouped feature profiles (right, z-score within feature). High-denial archetypes consistently pair elevated pressure with fast lane closure and compact support geometry; pressure alone is insufficient without the closure component.

Tight Squeeze is the highest-denial archetype (62.4% combined): it pairs strong carrier pressure with compressed F2/F3 support that removes secondary exits. **Heavy Disrupt** produces sustained pressure at a wider spread, yielding strong forced-dump rates. **Light Pressure** is the most permissive, with low pressure, slow closure, and the highest clean-exit rate (52.9%), making it the clearest coaching target for improvement. **Spread Support** sits in the middle: reasonable pressure but incomplete lane removal. The topology phase-space (Figure 6, right) reduces the picture to two coaching questions: how open is the best exit lane at the start, and how quickly does that lane close? The highest-denial structures occupy the *suffocating* quadrant, where an already-limited option gets tighter within the critical 3-second window.

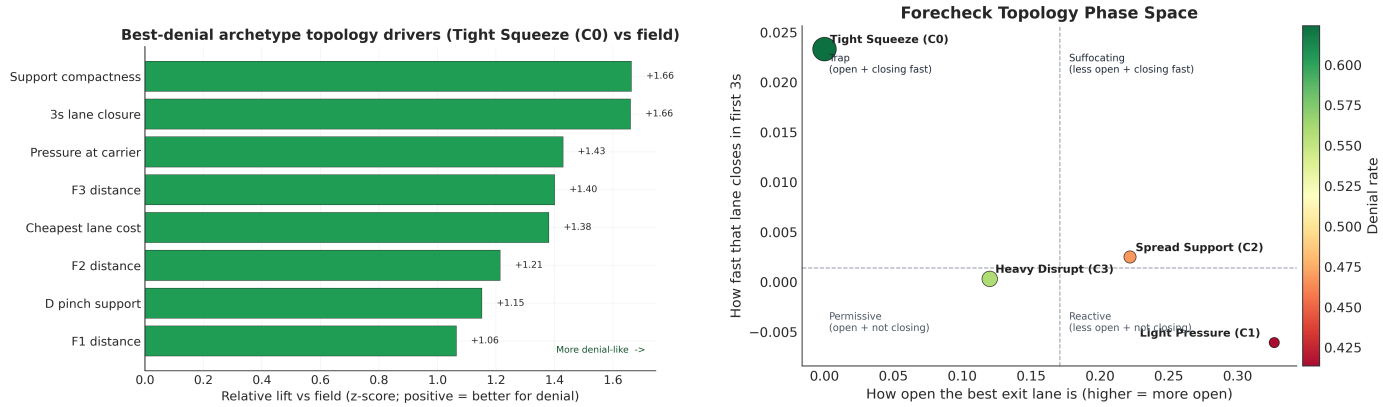

Figure 6: Relative lift profile for the best denial archetype (left; rightward bars = more denial-like relative to field average) and topology phase-space (right; quadrant labels reflect coach-language characterizations). High-denial structures occupy the "suffocating" quadrant: less open and closing fast.

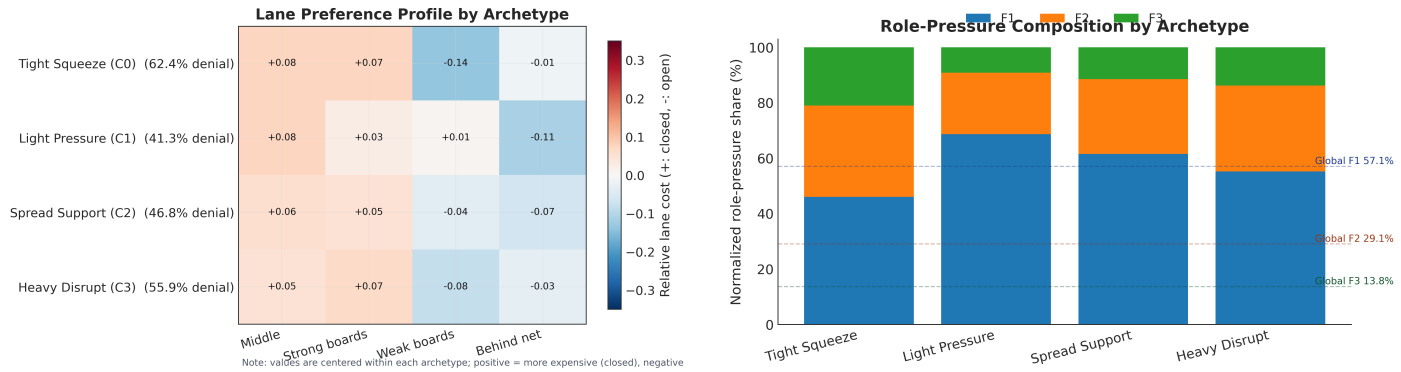

Figure 7: Lane preference profiles by archetype (left; positive = more closed than archetype mean) and role-pressure composition by archetype (right, with global F1/F2/F3 reference lines). High-denial archetypes close strong-side lanes first and preserve secondary-layer contribution integrity across F2 and F3.

Global normalized role composition is approximately F1 59.6%, F2 27.8%, F3 12.6%. High-denial archetypes maintain stronger F2 and F3 contribution relative to the field; the first forechecker alone cannot manufacture denial, because denial depends on coordinated layering from the second and third waves.

Coaching takeaway: improving only F1 pressure will understate the true ceiling. Denial rises most when F2 closes the secondary outlet within 2–3 seconds and F3 removes the behind-net release, allowing pressure to compound across layers rather than relying on a single forechecker.

4.4 Score-State Context, Closure Dynamics, and Predictive Performance

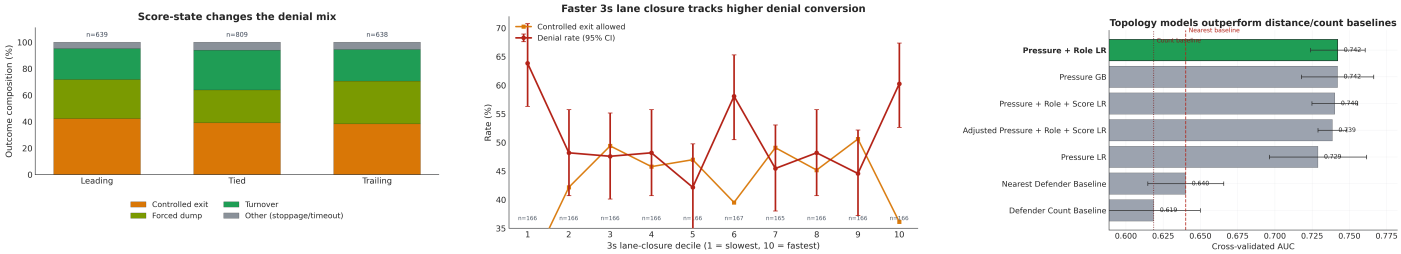

Figure 8: Score-state denial composition (left); closure-response by decile with 95% CI (center); model AUC comparison (right, best method highlighted). Trailing teams show higher forced-dump rates, consistent with desperation pressure that sacrifices structure. Faster positive 3-second closure tracks higher denial, and topology models outperform distance/count baselines.

Table 3: Turnover Prediction Performance (Primary Supervised Endpoint)

Model	AUC Mean	AUC Std	n Obs
Pressure + role logistic	0.742	0.018	1,569
Pressure gradient boosting	0.742	0.024	1,660
Pressure + role + score logistic	0.740	0.015	1,569
Adjusted pressure + role + score logistic	0.739	0.010	1,569
Pressure-only logistic	0.729	0.033	1,660
Nearest-defender baseline	0.640	0.025	1,665
Defender-count baseline	0.619	0.031	1,665

The best model in Table 3 is the pressure+role logistic model at 0.742 AUC. Relative to the nearest-defender baseline (0.640), this is a +0.102 absolute lift, or roughly a 16% improvement on the AUC scale. The smaller gain from 0.729 to 0.742 shows that **role-layer geometry** adds useful signal beyond total pressure alone: knowing where the second and third forecheckers are positioned improves prediction, not just knowing that pressure exists. Adding score state changes performance only negligibly (0.742 to 0.740), suggesting the spatial pressure structure is the primary driver.

4.5 Team Process Benchmarking

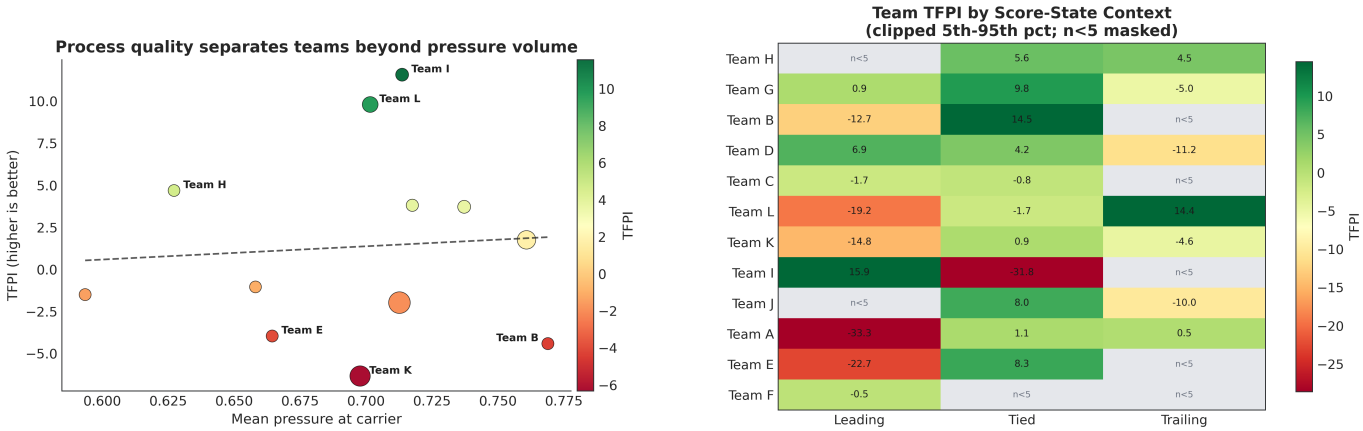

Figure 9: Team Forecheck Pressure Index (TFPI) overall (left, bubble size = retrieval count) and by score-state (right, clipped 5th–95th percentile, sparse cells masked). TFPI is a process metric; small sample caution applies at the team level across this 10-game dataset.

Team-level TFPI (**Team Forecheck Pressure Index**) is defined as $\text{TFPI} = 100 \cdot (\text{TurnoverRate} + 0.5 \cdot \text{ForcedDumpRate} - \text{ControlledExitRate})$. Higher TFPI is better: it rewards retrievals that end in turnovers, gives partial credit to forced dump-outs, and penalizes clean exits against. We use a 0.5 weight because a forced dump-out is meaningfully better than allowing a controlled exit, but not as valuable as winning the puck outright with a turnover. Team I (11.6) and Team L (9.8) lead; Team K (−6.3), Team B (−4.4), and Team E (−4.0) sit lowest.

5 Discussion and Operational Deployment

The central empirical finding is that denial is driven by *lane-removal geometry*, not nearest-defender distance alone. Three action points follow. First, coaches should evaluate retrieval pressure by asking whether the carrier's best lane became harder within the first three seconds, not just whether F1 arrived on time. Second, layered support matters: the best denial outcomes occur when F2 and F3 remove secondary exits quickly enough to trap the carrier after first contact. Third, team benchmarking should focus on process quality, because the topology signal remains predictive even when score state is added and has only negligible incremental value.

In practice, this supports a clip-review workflow organized around (1) which corridor the carrier identified as cheapest at retrieval start, (2) whether that lane closed within three seconds, and (3) whether secondary support and D-pinch help arrived in time. Because all signals are computed per episode from tracking, the system supports line-matching, role accountability, and pre-scouting. A natural extension is to weight denial events by downstream transition value so that future versions of RDT distinguish between "good" dump-outs and truly damaging turnovers on a common value scale.

6 Limitations

Four constraints apply: role labels are inferred from positional proximity (nearest offensive skater to puck = carrier) and require manual video audit for confirmation; this inference is most susceptible to error in screening and tight-cycling sequences where multiple offensive players converge simultaneously, though such situations are less prevalent in the confined D-zone retrieval episodes studied here than in open-play cycling contexts, and the expected misclassification rate is modest relative to the episode sample. Acceleration and heading estimates inherit noise from the underlying tracking system. Corridor costs treat lane difficulty in a common pressure-based currency, but the four exits are not equal in downstream value; a future extension should weight lanes by the transition value they typically create when completed. Episode segmentation is rule-based rather than learned from tagged possession chains. The supervised endpoint is event occurrence rather than downstream expected-goal or transition-value outcome.

7 Conclusion

Forechecking success on retrievals is better modeled as a continuous topology of pressure, corridor availability, and closure dynamics than as a discrete system label. Retrieval Denial Topology produces interpretable coaching signals and improves turnover prediction over standard distance and count baselines. The practical message is simple: strong forechecks are layered forechecks, and the decisive window is the first three seconds after retrieval.